Zetian Wu

+1-(443)-630-2430 | ztwu.nil@gmail.com | neal-ztwu.github.io

Education

Oregon State University — Corvallis, OR, USA
PhD in CS & AI (Dual-major with OSP Fellowship) — *Sept. 2022 – 2027*
Advisor: Prof. Liang Huang; Co-advisor: Prof. Stefan Lee

Johns Hopkins University — Baltimore, MD, USA
MSE in Data Science — *Jan. 2020 – Dec. 2021*

Zhejiang University — Hangzhou, China
BS in Physics | Minor in Finance — *Sept. 2015 – Jun. 2019*

Research Experience

Research Assistant — Sept. 2022 – present
with Prof. Liang Huang, Prof. Stefan Lee — *OSU, USA*

Simultaneous Translation between Sign Languages [1] (To NAACL 2027)
- **Problem**: Generated target-sign motion before full-source availability for streaming ASL/CSL/DGS translation.
- **Techniques**: Combined mBART-large-cc25 with a frozen VQ-VAE over SMPL-X poses; implemented prefix-to-prefix training, test-time wait-k decoding, stochastic multi-path supervision, and CA-Stream-AL.
- **Results**: Cut computation-aware latency by 38% versus full-sentence decoding, with 9% higher DTW-PA-MPJPE and 2.1-point lower BLEU-4.

Direct Translation between Sign Languages [2] (Submitted to ICLR 2027)
- **Problem**: Replaced three-stage spoken-text cascades with direct ASL/CSL/DGS inference.
- **Techniques**: Created 112K cross-lingual pairs from How2Sign, CSL-Daily, and PHOENIX-2014T via TranslateGemma-4B back-translation; jointly trained mBART-large-cc25 for T2S/S2S with a frozen VQ-VAE, and curated an evaluation subset using Qwen3-8B, multilingual Sentence-BERT, and bilingual review.
- **Results**: On held-out BT-input data, beat cascades in all six directions: mean DTW-PA-MPJPE -19%, back-translation BLEU-4 +50%, and single-GPU inference 2.3× faster.

Geometry-Aware Text-to-Sign Generation [3]
- **Problem**: Generated variable-length 3D sign poses from text with articulated hands and consistent skeletons.
- **Techniques**: Used XLM-R with OpenPose, 2D-to-3D lifting, inverse kinematics, and skeleton normalization; added parent-relative and bone-length reweighting, bone-length/bone-pose losses, and EOS prediction.
- **Results**: Reached 14.22 back-translation BLEU-4 on PHOENIX14T dev (+3.08 points over XLM-R) and reduced bone-length and movement-variance discrepancies by 18.76% and 5.48%.

Research Assistant — Jun. 2020 – Apr. 2022
with Prof. João Sedoc, Prof. Lyle Ungar — *NYU/UPenn, USA*

Inducing Generalizable and Interpretable Lexica [4]
- **Problem**: Studied how model architectures and attribution methods shape induced psycholinguistic lexica.
- **Techniques**: Trained SVM/FFN on averaged FastText embeddings and fine-tuned RoBERTa/DistilBERT; induced lexica using single-token importance, masking, and Partition SHAP.
- **Results**: Context-oblivious models yielded more generalizable, interpretable lexica than context-aware ones.

Research Assistant — Mar. 2021 – Sep. 2021
with Prof. Louis-Philippe Morency — *CMU, USA*

MultiBench: Multiscale Benchmarks for Multimodal Representation Learning [5]
- **Problem**: Benchmarked multimodal generalization, efficiency, and robustness across diverse tasks and modalities.
- **Techniques**: Contributed early/late fusion, CCA, and MFM to MultiBench's modular PyTorch pipeline.
- **Results**: Project-wide: 15 datasets/10 modalities/20 tasks; method transfer improved SOTA on 9/15 datasets.

Research Assistant — Apr. 2018 – Aug. 2019
Intelligent Computing & System Lab, with Prof. Qinming He — *ZJU, China*

Interactive Rare-Category-of-Interest Mining from Large Datasets [6]
- **Problem**: Enabled real-time discovery of rare, user-relevant classes in large datasets.
- **Techniques**: Built RCD with offline knowledge abstraction and RCE with collaborative reconstruction.
- **Results**: Reduced per-query complexity from $O(n^2)$ to $O(\log n)$; improved RCE accuracy 11.75% over prior SOTA.

Work Experience

Research Scientist Intern June 2024 – Sept. 2025
Genies *San Mateo, CA, USA*

- Built an LLM user-insight application over profiles, app logs, and chat histories; designed privacy-aware data normalization for behavioral analytics.
- Explored GRPO post-training and MCTS-guided reasoning for efficient text-to-SQL generation while preserving overall generation accuracy.

Applied Scientist Intern July 2023 – Oct. 2023
Amazon *San Francisco, CA, USA*

- Researched Transformer-based generative models for long multimodal sequences, exploring architecture and training designs for cross-modal dependencies and long-context coherence.

Applied Scientist Intern May 2022 – Aug. 2022
Amazon *Seattle, WA, USA*

- Built FMDA on frozen XLM-R-base to disentangle language-specific, template-semantic, and residual-semantic representations using language-ID, reconstruction, and contrastive objectives.
- On MASSIVE, consistently outperformed XLM-R and a strong language-disentanglement baseline on monolingual template and cross-lingual semantic retrieval across EN–DE, EN–ES, and EN–JA [7].

Machine Learning Engineer Aug. 2019 – Mar. 2020
Hangzhou Enjoymusic Technology Co. Ltd. *Hangzhou, China*

- Built Transformer-XL/discriminator models for music style transfer and a VAE for drum-conditioned MIDI generation; ported TypeScript MIDI tooling to Python for product integration.

Publications and Preprints

[1] **Zetian Wu**, Bowen Xie, Stefan Lee, and Liang Huang. *Simultaneous Translation between Sign Languages*. 2026. arXiv: 2609.35608 [cs.CV].

[2] **Zetian Wu**, Bowen Xie, Wuyang Meng, Milan Gautam, Stefan Lee, and Liang Huang. *Direct Translation between Sign Languages*. 2026. arXiv: 2605.20588 [cs.CL].

[3] **Zetian Wu**, Tianshuo Zhou, Stefan Lee, and Liang Huang. *Geometry-Aware Losses for Structure-Preserving Text-to-Sign Language Generation*. 2025. arXiv: 2509.23011 [cs.CV].

[4] *Yilin Geng, ***Zetian Wu**, Roshan Santhosh, Tejas Srivastava, Lyle Ungar, and João Sedoc. *Inducing Generalizable and Interpretable Lexica*. In: *Findings of the Association for Computational Linguistics: EMNLP 2022*. Abu Dhabi, United Arab Emirates: Association for Computational Linguistics, Dec. 2022.

[5] Paul Pu Liang, Yiwei Lyu, Xiang Fan, **Zetian Wu**, Yun Cheng, Jason Wu, Leslie Yufan Chen, Peter Wu, Michelle A Lee, Yuke Zhu, Russ Salakhutdinov, and Louis-Philippe Morency. *MultiBench: Multiscale Benchmarks for Multimodal Representation Learning*. In: *Thirty-fifth Conference on Neural Information Processing Systems Datasets and Benchmarks Track (Round 1)*. 2021.

[6] Zhenguang Liu, Sihao Hu, Yifang Yin, Jianhai Chen, Kevin Chiew, Luming Zhang, and **Zetian Wu**. *Interactive Rare-Category-of-Interest Mining from Large Datasets*. In: *Proceedings of the AAAI Conference on Artificial Intelligence* 34.04 (Apr. 2020).

[7] **Zetian Wu**, Zhongkai Sun, Zhengyang Zhao, Sixing Lu, Chengyuan Ma, and Chenlei Guo. *Fine-grained Multi-lingual Disentangled Autoencoder for Language-agnostic Representation Learning*. In: *Proceedings of the Massively Multilingual Natural Language Understanding Workshop (MMNLU-22)*. Abu Dhabi, United Arab Emirates (Hybrid): Association for Computational Linguistics, Dec. 2022.

Skills and Additional Information

Programming & Frameworks: Python, C/C++, SQL, Linux, Git, PyTorch, Hugging Face Transformers, scikit-learn, NumPy, pandas, multi-GPU training
LLMs & Generative AI: LLM fine-tuning/post-training (GRPO), MCTS reasoning, LLM evaluation, text-to-SQL, synthetic data/back-translation, data curation, multilingual/long-context modeling, Sentence-BERT
Deep Learning & Multimodal: Transformers, seq2seq/multi-task learning, VQ-VAE/VAE, contrastive/few-shot learning, multimodal fusion, sign-language/text-to-motion generation, OpenPose, SMPL-X, inverse kinematics, SHAP
Awards & Honors: China Collegiate Computing Contest–AI Innovation Top Prize; COMAP Honorable Mention; OSU Outstanding Scholar; ZJU Top-notch Talent Scholarship